

Semi-supervised Dual-Branch Network for image classification

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ABSTRACT

In this work, we reveal an essential problem rarely discussed in current semi-supervised learning literatures: the learned feature distribution mismatch problem between labeled samples and unlabeled samples. It is common knowledge that learning from the limited labeled data easily leads to overfitting. However, the difference between the inferred labels of unlabeled data and the ground truths of labeled data may make the learned features of labeled and unlabeled data have different distributions. This distribution mismatch problem may destroy the assumption of smoothness widely used in semi-supervised field, resulting in unsatisfactory performance. In this paper, we propose a novel Semi-supervised Dual-Branch Network (SDB-Net), in which the first branch is trained with labeled and unlabeled data, and the other is trained with the predictions of unlabeled data generated from the first branch only. To avoid the different distributions between ground-truth labels and inferred labels for the unlabeled data, we proposed an effective co-consistency loss to overcome the mismatch problem and a mix-consistency loss to make each branch learn a consistent feature representation. Meanwhile, we designed an augmentation supervised loss for the first branch to further alleviate the mismatch problem. With the designed three kinds of losses, the proposed SDB-Net can be efficiently trained. The experimental results on three benchmark datasets, such as CIFAR-10, CIFAR-100 and SVHN, show the superior performance of the proposed SDB-Net.

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1. Introduction

In recent years, deep learning has achieved remarkable performance in various supervised learning tasks, such as image classification [1] and object detection [2]. As neural network models [3–5] are developed deeper and deeper with hundreds or thousands of layers, a large amount of labeled training data are required. However, due to the time and labor costs as well as the limits of expertise, a large-scale labeled dataset is a scarce resource in many machine learning applications, such as few-shot image classification [6,7] and medical image analysis tasks [8–10]. In comparison, with the rapid developments of mobile phones and camera sensors, a large amount of data without labels could be obtained easily and cheaply. Consequently, how to train an effective deep learning model from abundant unlabeled data and limited labeled data has been paid great attention to.

An attractive approach to address the above issue is semi-supervised learning [11], which aims to learn a better model

by using all the data than that learned by supervised learning with only limited labeled data alone. Semi-supervised learning has attracted much attention and many approaches have been proposed. Liu et al. [12] proposed a rough set based semi-supervised feature selection to predict the unlabeled data. Semi-supervised support vector machine [13,14] employed the unlabeled data to regularize the decision boundary so that it can pass through low-density regions of sample space. Semi-supervised support vector clustering [15] extended the clustering to the semi-supervised field by introducing a fuzzy Laplacian matrix. Co-training method [16] typically involved two learners to improve them through the exploitation of their disagreements. The key idea of many semi-supervised learning methods is the smoothness assumption, i.e., data points close to each other are likely to have the same label. The foundation behind the smoothness assumption is that data belonging to the same class should have the same data distribution whether or not they are labeled. Based on this assumption, many works have been proposed, such as label propagation [17], graph regularization [18], cluster algorithm [19], manifold-based method [20,21]. For instance, graph-based semi-supervised methods [17,18] encoded both the labeled data and unlabeled data in a graph and then estimated the class of unlabeled data with the similarities between labeled data and unlabeled data. However, traditional semi-supervised learning

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methods, which heavily rely on the handcrafted and off-the-shelf features, lack enough ability to explore complex natural data from the practical applications.

Inspired by the success of supervised deep learning, training deep learning models in a semi-supervised fashion gains an increasing interest. Semi-supervised deep learning can seek to alleviate the need of large amounts of labeled data by leveraging unlabeled data in an unsupervised way. A popular way [22,23] in semi-supervised deep learning is to pre-train a deep network with unlabeled data for a good initialization, followed by fine-tuning with labeled data in a supervised fashion. Rasmus et al. [24] proposed a novel network architecture, LadderNet, which combined the supervised and unsupervised learning in a denoising autoencoder by the skip connection. Rifai et al. [25] trained a contrastive auto-encoders (CAEs) to learn a data manifold and forced the model approximately invariant to local perturbations along the manifold. Although above methods [24,25,22,23] can simultaneously apply both supervised and unsupervised learning, the unsupervised learning of unlabeled data is independent of supervised learning, having not enough capacity to make full use of unlabeled data. Other methods [26,27] proposed to train the deep network with unlabeled data by self-training, in which the network was trained with the limited labeled data and the pseudo labels of unlabeled data were estimated based on the extracted features. Lee et al. [28] proposed to train the deep network by regarding the predicted class of unlabeled data as their labels. Chen et al. [29] utilized an external memory module to construct the pseudo targets of unlabeled data. Although above methods [26–29] employed the unlabeled data by combining the supervised learning and self-label technique, they did not design a unified objective loss function to jointly use the labeled and unlabeled data for the classification task, resulting in that the discriminations hidden in the unlabeled data are not well exploited.

Recent consistency-regularization methods [30–32] have achieved the state-of-the-art performances in the classification tasks. The consistency regularization forces the model to output the same prediction for the same input under different small perturbations (e.g., gaussian noise and Dropout [33]), encouraging the model to be smooth around each single data point. However, the consistency-regularization methods only focus on the constraint of each single data point, while ignore the relationships between data points.

The above consistency strategies for improving the deep model performance are built on the semi-supervised smoothness assumption, which may not hold always. On the one hand, the feature representations of labeled data are learned with ground truths, while those of unlabeled data are learned with inferred pseudo labels which usually contain incorrect predictions. The difference between the distributions of ground truths and inferred labels guide the model to learn different feature representation distributions. On the other hand, the overfitting due to the limited labeled data aggravates the distribution mismatch problem. Feature representations of Pseudo-Label [28] and our proposed method have been shown by visualizing input vectors of the last fully-connected layer in the experiments in Section 4.3. To show the distribution mismatch, we visualize 4000 labeled samples and 4000 randomly selected unlabeled samples from CIFAR-10. For better visualization, we use t-SNE [34] to generate the resulting 2-D deep features. The results are shown in Fig. 1. Specifically, as shown in Figs. 1(a–c), the learned feature representation distributions of labeled and unlabeled data may mismatch, which hurts the smoothness assumption and leads to little improvements additionally brought by unlabeled data.

To solve the above issues, in the paper, we propose a novel network architecture (as shown in Fig. 2), the Semi-supervised

Dual-Branch Network (SDB-Net) that consists of a shared module and two branches with the same structure, to address the learned feature distribution mismatch problem between labeled and unlabeled data. The first branch is trained with labeled and unlabeled data with our designed consistency-regularizations, while the second branch imitates the predictions of unlabeled data from the first branch, avoiding the different distributions of ground truths and inferred labels. With the shared module, the proposed SDB-Net would learn the matching feature representations from the second branch and feed back to the first branch. We demonstrate the effectiveness of our SDB-Net through a feature visualization experiment and extensive comparisons with the state-of-the-art semi-supervised methods on the benchmark datasets (e.g., CIFAR10, CIFAR-100 and SVHN).

Our contributions are summarized as follows:

- We present a rarely discussed learned feature distribution mismatch issue between labeled data and unlabeled data, which is harmful to semi-supervised smoothness assumption.
- We design a dual-branch architecture to explicitly address the learned feature distribution mismatch problem and a co-consistency regularization to train the dual-branch architecture efficiently.
- As for preventing overfitting due to the limited labeled data, we design an augmentation supervised loss by adopting a simple mixup [35] variant in labeled data.

2. Related work

In this section, we will introduce existing methods related to semi-supervised learning. Among wide literatures on semi-supervised learning techniques, we focus mainly on those which are currently state-of-the-art. In the following, we will refer to the related researches about the smoothness assumption, semi-supervised deep learning, and the distribution mismatch problem.

2.1. Assumption of semi-supervised learning

The key idea of most semi-supervised learning problems is the prior assumption of smoothness, which means: (i) data points close to each other are likely to have the same label; (ii) data points with the same structure (e.g., a cluster or a manifold structure) are likely to have the same label.

The main difference among these semi-supervised learning methods, such as graph-based methods [18,17], cluster algorithm [19], manifold-based method [20], is the way in which they realize the smoothness assumption. For semi-supervised deep learning methods, the smoothness assumption is applied in the intermediate representations of neural network. However, the smoothness assumption may not hold always, especially when the deep model learns different feature representation distributions between labeled and unlabeled data due to different supervised signals (e.g., ground truths of labeled data and inferred pseudo labels of unlabeled data).

2.2. Semi-supervised deep learning

Most work related to our method are the semi-supervised deep learning models with consistency regularization, which prefers a stable estimation for an input. In the learning of the target predictions for the same samples, consistency regularization encourages model to be smooth around data points and makes model invariant to small perturbations. IT model [31] is a simple method which forces the learned classifier to make the same prediction for various generated samples of an input by

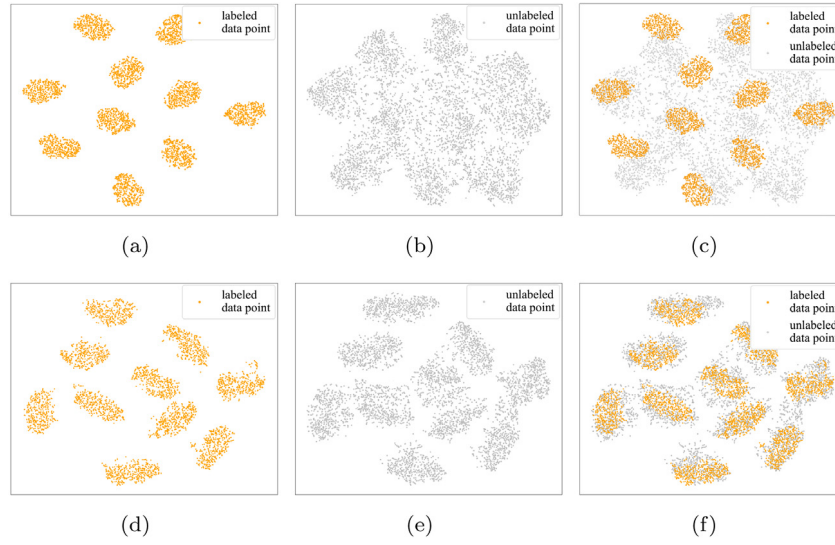


Fig. 1. Feature visualizations of the semi-supervised baseline, Pseudo-Label, and our SDB-Net. The top Figs.(a–c) are the learned feature representations of Pseudo-Label. The bottom Figs.(d–f) are the learned features of SDB-Net. The orange points denote the learned feature representations of labeled data points. The silver points denote the learned feature representations of unlabeled data. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

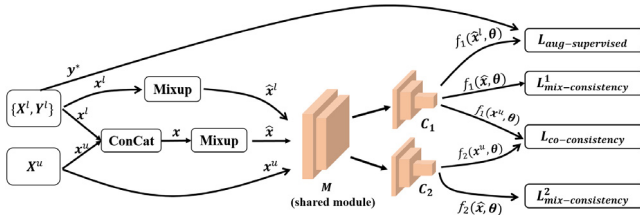


Fig. 2. The semi-supervised dual-branch network.

data augmentation. However, the Π model uses its prediction of the augmented sample as the target prediction, which makes the model training unstable and convergence slowly. Tarvainen et al. [30] proposed a teacher–student framework, MeanTeacher, using an exponential moving average of student model’s weights to construct a teacher model, which can generate a stable target prediction. The teacher model supervises the student model learning via its more stable and accurate target predictions, while the student benefits to the teacher via exponential moving averaging of its outputs, which forms a virtuous feedback cycle. For further improving the targets of unlabeled data, Li et al. [36] proposed certainty-driven consistency loss (CCL) to dynamically select the confident predictions and Verma et al. [37] employed recent start-of-the-art data augmentation to enhance the teacher model.

Instead of considering the smoothness around single data point only, Luo et al. [38] proposed a method, named Smooth Neighbors on Teacher Graphs (SNTG). In the SNTG, the local connection information between data points in a mini-batch is explored and the smoothness on the data manifold is introduced through a teacher graph. With a transductive label propagation framework, Iscen et al. [39] constructed a nearest neighbor graph on the embeddings of all data points to learn a global manifold structure. Shi et al. [40] proposed the Min–Max Feature (MMF) regularization to improve inter-class separability and intra-class compactness. Focusing on the perturbations, Miyato et al. [32] proposed Virtual Adversarial Training (VAT) to approximate an adversarial perturbation which can significantly affect prediction probability distribution of classifier. Instead of using L_p -norm of

the Jacobian matrix in VAT, Yu et al. [41] proposed an extension of VAT, tangent-normal adversarial regularization (TNAR), by applying VAT along and orthogonal to the tangent space of the data manifold.

While above consistency-regularization-based methods achieve promising performances on the benchmark datasets, they have not discovered or focused on the learned feature distribution mismatch issue. The mismatch problem is attributed to the difference between the inferred labels of unlabeled data and the ground truths of labeled data, and is also closely associated with the overfitting due to the limited labeled data. A new network architecture and appropriate regularizations, which are different from previous models, are needed to reduce the distribution mismatch of learned feature representations.

2.3. Learned feature distribution mismatch

Learned feature distribution mismatch, which was rarely discussed in the literatures, is a new perspective of semi-supervised deep learning. In the semantic space, the distributions between labeled data and unlabeled data that come from the same class are the same. However, in the learned feature space, the distributions between labeled data and unlabeled data are biased to be different as the deep learning model is trained with limited ground truth labels and many inferred pseudo labels. Here the learned feature distribution mismatch we focus on is different from the domain adaptation. The latter comes from two different datasets which have different distributions; however, the former is raised between labeled data and unlabeled data in the same dataset.

Wang et al. [42] formulated a sampling bias issue which causes empirical distribution mismatch between labeled and unlabeled data, and proposed an adversarial training strategy to minimize the distribution distance. Zajc et al. [43] proposed to compute batch-normalization statistics separately for the labeled and unlabeled data to alleviate the class mismatch problem and the feature distributions mismatch problem. Jackson and Schulman [44] proposed the label gradient alignment (LGA) to define a semantically meaningful distance metric on the input data and used the resulting metric to compute labels for the unlabeled data.

While above methods implicitly alleviate the distribution mismatch between learned feature representations of labeled and

unlabeled data, they usually rely on the sophisticated Generative Adversarial Networks (GAN) training process [42] or the complex computing metric [44], which make the optimization and convergence difficult. Moreover, lacking of explicit objective for tackling the distribution mismatch problem makes them inefficient to exploit the relationship of the learned feature distributions between labeled data and unlabeled data.

3. Method

To address the learned feature distribution mismatch problem, we design a novel semi-supervised dual branch network (SDB-Net), in which labeled and unlabeled feature distributions are jointly learned by these two branches. The first branch is trained with labeled and unlabeled data, and the second branch is learned with the predictions of unlabeled data computed from the first branch to avoid mismatch problem. To collaboratively explore the discrimination of labeled and unlabeled data, we propose a simple yet effective co-consistency loss, which can reduce the learned feature mismatch issue between labeled data and unlabeled data. With constructing elaborate perturbations essential for the consistency regularization, we propose a mix-consistency regularization to boost the performance. Moreover, for preventing from overfitting, we propose an augmentation supervised loss with the popular data augmentation to further alleviate the mismatch problem.

In this section, we design the structure of the proposed semi-supervised dual-branch network (SDB-Net), which uses two branches to separate the effect caused by the ground truths of labeled data and the inferred labels of unlabeled data. As shown in Fig. 2, SDB-Net is a deep neural network which consists of three components: a shared module M , which maps from images to a latent space for feature learning, and two branch modules C_1 and C_2 , which jointly map from the latent space to the class space for predictions. The first branch (denoted by C_1 without loss of generality) is trained with a popular semi-supervised deep learning method and the C_2 branch is trained with inferred labels of unlabeled data to avoid the mismatch problem. Then the C_1 branch gets the matching feedback from the learned feature through the shared module M .

The proposed SDB-Net that unifies all these terms into one framework is formed as follow:

$$L = L_{\text{aug-supervised}} + L_{\text{co-consistency}} + w_1 L_{\text{mix-consistency}}^1 + w_2 L_{\text{mix-consistency}}^2 \quad (1)$$

where $L_{\text{aug-supervised}}$ is the augmentation supervised loss of the limited labeled data in the C_1 branch to prevent overfitting, $L_{\text{co-consistency}}$ is the co-consistency loss to reduce the mismatch issue between labeled and unlabeled features, and $L_{\text{mix-consistency}}^1$ and $L_{\text{mix-consistency}}^2$ are the consistency-based losses of the C_1 branch and of the C_2 branch, respectively. The weights w_1 and w_2 are the hyperparameters to control the importances of mix-consistency regularizations. Following [30], w_1 and w_2 ramp up, starting from zero, along a Gaussian curve, to the final values during training.

In this following, we first illustrate our co-consistency loss. Then we present mix-consistency regularization and augmentation supervised loss which can alleviate the mismatch problem.

3.1. Co-consistency loss

As previously mentioned, the popular semi-supervised deep learning (e.g., consistency regularization methods) will cause the learned feature mismatch problem, which mainly results from the difference between ground truths of labeled data and inferred labels of unlabeled data.

The proposed SDB-Net designs two branches to effectively learn discrimination from labeled and unlabeled data. Typically, the C_2 branch learns unlabeled feature representation under the supervision of its predicted class outputted by the C_1 branch. In order to prevent the learned feature distribution mismatch issue in C_2 branch, we proposed the co-consistency loss to train the C_2 branch effectively. Given an unlabeled sample $\mathbf{x}^u \in \mathbf{X}^u$, where $\mathbf{X}^u$ represents the unlabeled data in the mini-batch, the co-consistency loss can be formulated as follows.

$$L_{\text{co-consistency}} = \frac{1}{B^u} \sum_{i=1}^{B^u} \|f_2(\mathbf{x}_i^u, \theta) - f_1(\mathbf{x}_i^u, \theta)\|_2^2 \quad (2)$$

where B^u is the sample number of $\mathbf{X}^u$ in mini-batch and θ is the learnable parameters of network. $f_1(\mathbf{x}_i^u, \theta)$ is the output of C_1 branch and $f_2(\mathbf{x}_i^u, \theta)$ is the output of C_2 branch. It is noted that the former is the target of the latter and the gradient propagation is carried out in C_2 branch only.

The co-consistency loss can be viewed as a form of Knowledge Distillation or Mutual Learning, which has been verified to be effective to train a compact yet accurate model. Since the C_1 branch is slightly overfitting due to the limited labeled data, the C_2 branch is trained with the predictions of unlabeled data from the C_1 branch only.

3.2. Mix-consistency regularization

By interpolating convex combinations of sample pairs and corresponding target vectors into original data space and label space, respectively, recent data augmentation method, mixup [35], achieved the state-of-the-art performance in supervised visual tasks. Here, we apply the idea of Mixup to consistency regularization and adapt it to the SDB-Net.

Given a sample $\mathbf{x}_p$, we select randomly a sample $\mathbf{x}_q$ in a mini batch to construct a sample pair. For $\mathbf{x}_p$, let $\mathbf{y}_p^1 = f_1(\mathbf{x}_p, \theta')$ and $\mathbf{y}_p^2 = f_2(\mathbf{x}_p, \theta')$ be the corresponding target vectors from the C_1 and C_2 branches, respectively, where the θ' is a moving average of θ . Analogously, $\mathbf{y}_q^1 = f_1(\mathbf{x}_q, \theta')$ and $\mathbf{y}_q^2 = f_2(\mathbf{x}_q, \theta')$ are the target vectors of $\mathbf{x}_q$ from the C_1 and C_2 branches, respectively. The Mixup operation of SDB-Net is defined as:

$$\hat{\mathbf{x}} = \lambda \mathbf{x}_p + (1 - \lambda) \mathbf{x}_q \quad (3)$$

$$\hat{\mathbf{y}}^1 = \lambda \mathbf{y}_p^1 + (1 - \lambda) \mathbf{y}_q^1$$

$$\hat{\mathbf{y}}^2 = \lambda \mathbf{y}_p^2 + (1 - \lambda) \mathbf{y}_q^2$$

where $\hat{\mathbf{y}}^1$ and $\hat{\mathbf{y}}^2$ are the mixup targets for C_1 and C_2 branches, respectively. And $\lambda \in [0, 1]$ is sampled from an prior beta distribution $\beta(\alpha, \alpha)$, where α is a hyperparameter.

Denote $L_{\text{mix-consistency}}^1$ and $L_{\text{mix-consistency}}^2$ the mix-consistency regularizations for C_1 and C_2 branches, respectively. They can be written as:

$$L_{\text{mix-consistency}}^1 = \frac{1}{B} \sum_{i=1}^B \|f_1(\hat{\mathbf{x}}_i, \theta) - \hat{\mathbf{y}}_i^1\|_2^2 \quad (4)$$

$$L_{\text{mix-consistency}}^2 = \frac{1}{B} \sum_{i=1}^B \|f_2(\hat{\mathbf{x}}_i, \theta) - \hat{\mathbf{y}}_i^2\|_2^2 \quad (5)$$

where B is the mini-batch size.

The combination of Mixup and consistency regularization can boost the performance and alleviate the learned feature mismatch. On one hand, the data augmentation is essential for consistency regularization. Mixup is a good perturbation for consistency regularization since it can greatly transform the data while keep the underlying information. On the other hand, learning from mixup inputs and mixup targets prevents the network

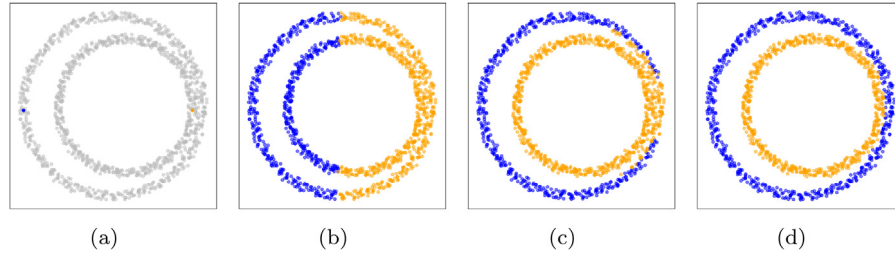


Fig. 3. The role of the mix-consistency regularization. (a) The silver points denote unlabeled data. The orange point and blue point denote labeled data points from two classes. (b) The result of semi-supervised baseline, Pseudo-Label. (c) The result of semi-supervised baseline with consistency regularization. (d) The result of semi-supervised baseline with mix-consistency regularization. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1
CNN-13 architecture.

Block	Layer	Hyperparameters
1st block	Convolutional	128 filters, 3×3 , pad = 'same', LReLU($\alpha = 0.1$)
	Convolutional	128 filters, 3×3 , pad = 'same', LReLU($\alpha = 0.1$)
	Convolutional	128 filters, 3×3 , pad = 'same', LReLU($\alpha = 0.1$)
	Pooling	2×2 Maxpool
2nd block	Convolutional	256 filters, 3×3 , pad = 'same', LReLU($\alpha = 0.1$)
	Convolutional	256 filters, 3×3 , pad = 'same', LReLU($\alpha = 0.1$)
	Convolutional	256 filters, 3×3 , pad = 'same', LReLU($\alpha = 0.1$)
	Pooling	2×2 Maxpool
3rd block	Convolutional	512 filters, 3×3 , pad = 'valid', LReLU($\alpha = 0.1$)
	Convolutional	256 filters, 1×1 , pad = 'same', LReLU($\alpha = 0.1$)
	Convolutional	128 filters, 1×1 , pad = 'same', LReLU($\alpha = 0.1$)
	Pooling	Average pool $\rightarrow 1 \times 1$
	Fully-connected Softmax	

from being misled to the learned feature distribution mismatch. Specifically, mixup inputs, generated between labeled data and unlabeled data, force the network to model the distributions lying between labeled and unlabeled data, which helps to bridge the gap between two learned feature distributions.

As depicted in Fig. 3, Fig. 3(a) presents a toy dataset of two circles in the semi-supervised setting, in which two data points from two classes are chosen as labeled data, with the remaining data as unlabeled data. The Pseudo-Label estimates the unlabeled data simply relying on the similarities between labeled data and unlabeled data. And the consistency regularization works on the manifold, which can capture the intrinsic structures embedded in the data. However, from Fig. 3(c), the consistency regularization cannot handle well the samples near the decision boundary. The mix-consistency regularization identifies correctly the samples near the decision boundary by identifying the distribution between labeled and unlabeled data, which can well model the data near the boundary as shown in Fig. 3(d).

3.3. Augmentation supervised loss

The other reason causing the learned feature distribution mismatch issue is the overfitting due to the limited labeled data. In semi-supervised learning the amount of unlabeled data is much larger than that of labeled data, since it is much easier to obtain unlabeled data. To make full use of a large-scale unlabeled data, a complex deep model, which is prone to fitting the limited labeled data very well (i.e., overfitting), is preferred.

To tackle this issue, we apply a data augmentation processing to labeled data and propose a mixup variant. Given a labeled data $(\mathbf{x}_p^l, \mathbf{y}_p^*) \in \{\mathbf{X}^l, \mathbf{Y}^l\}$, where $\mathbf{X}^l$ represents the labeled data in the mini-batch and $\mathbf{Y}^l$ represents the corresponding ground truth, we select randomly a labeled data $(\mathbf{x}_q^l, \mathbf{y}_q^*) \in \{\mathbf{X}^l, \mathbf{Y}^l\}$ in the mini-batch to construct a sample pair. Then the augmented data is

$$\hat{\mathbf{x}}_p^l = \lambda \mathbf{x}_p^l + (1 - \lambda) \mathbf{x}_q^l \quad (6)$$

where λ is sampled from the beta distribution $\beta(\alpha, \alpha)$. In a similar way we get its augmented label via $\hat{\mathbf{y}}_p^l = \mathbf{y}_p^* + (1 - \lambda) \mathbf{y}_q^*$.

Let $\rho(\mathbf{x}, \theta)$ is the softmax probabilities of $\mathbf{x}$, where $\mathbf{y}$ is the label vector of input $\mathbf{x}$ and A is the number of classes. The originally proposed MixUp loss can be written as follow

$$L_{aug-supervised} = \sum_{i=1}^{B^l} \text{CrossEntropy}(\hat{\mathbf{x}}_i^l, \hat{\mathbf{y}}_i^l) \quad (7)$$

where the popular cross entropy cost function, CorssEntropy, is defined as follow

$$\text{CrossEntropy}(\mathbf{x}, \mathbf{y}) = - \sum_i^A y_i \log \rho_i(\mathbf{x}, \theta) \quad (8)$$

The augmented label, $\hat{\mathbf{y}}_i^l$, represents the similarity of data labels; however, the similarity of data features may not be well preserved by Eq. (7) due to the complexity of its optimization (e.g., many local minimums). Thus we define our augmentation supervised loss as follow:

$$L_{aug-supervised} = \sum_{i=1, j=rand(i)}^{B^l} \{ \lambda \cdot \text{CrossEntropy}(\hat{\mathbf{x}}_i^l, \mathbf{y}_i^*) + (1 - \lambda) \cdot \text{CrossEntropy}(\hat{\mathbf{x}}_i^l, \mathbf{y}_j^*) \} \quad (9)$$

where $rand(i)$ is a function to randomly select a labeled sample for i th labeled sample and B^l is the number of $\mathbf{X}^l$ in mini-batch.

Ideally, the proposed loss $L_{aug-supervised}$ in Eq. (9) would lead to a better model to preserve the consistence of learned feature by trusting in progressively better labels during training. With interpolating linearly between labeled samples, augmentation data can enlarge the labeled dataset greatly and make the training process more stable and less risky for overfitting. Intuitively, the MixUp loss encourages the interpolated samples to be smooth to their interpolated labels, and the augmentation supervised loss forces model to correctly classify the input to a class.

3.4. Optimization

We have presented our semi-supervised dual-branch network (SDB-Net) with two branch modules and a shared module and illustrated how it works with the proposed co-consistency loss, mix-consistency regularizations, and augmentation supervised loss. The dual-branch network architecture, as shown in Fig. 2, is easy and efficient to implement. Specifically, we take the CNN-13 architecture, as shown in Table 1, as an example. We use the 1st and 2nd blocks to make up the shared module M and construct the branch modules C_1 and C_2 by duplicating the 3rd block. It is

easy to extend the popular convolution neural networks to the dual-branch architecture.

The entire training pipeline for the proposed SDB-Net is depicted in Algorithm 1. For each mini-batch, we first use the mixup interpolation strategy to generate mixup data via Eq. (3) and mixup labeled data via Eq. (6), which can be done during mini-batch data preparation. Then, the network takes them as input to compute the loss terms via Eqs. (2), (4), (5), and (9), of which the gradients are computed as follow.

The gradient of the Eq. (2) is

$$\nabla L_{co-consistency} = \frac{2}{B^u} \sum_{i=1}^{B^u} \{(f_2(\mathbf{x}_i^u, \theta) - f_1(\mathbf{x}_i^u, \theta)) \cdot \nabla f_2(\mathbf{x}_i^u, \theta)\} \quad (10)$$

where $\nabla f_2(\mathbf{x}_i^u, \theta)$ is the gradient of C_2 branch for input $\mathbf{x}_i^u$. It is worth noticing that the gradient is only carried out in C_2 branch.

The gradients of the Eqs. (4) and (5) are presented as follows.

$$\nabla L_{mix-consistency}^1 = \frac{2}{B} \sum_{i=1}^B \{(f_1(\hat{\mathbf{x}}_i, \theta) - \hat{\mathbf{y}}_i^1) \cdot \nabla f_1(\hat{\mathbf{x}}_i, \theta)\} \quad (11)$$

$$\nabla L_{mix-consistency}^2 = \frac{2}{B} \sum_{i=1}^B \{(f_2(\hat{\mathbf{x}}_i, \theta) - \hat{\mathbf{y}}_i^2) \cdot \nabla f_2(\hat{\mathbf{x}}_i, \theta)\} \quad (12)$$

where $\nabla f_1(\hat{\mathbf{x}}_i, \theta)$ and $\nabla f_2(\hat{\mathbf{x}}_i, \theta)$ are the gradients of C_1 branch and C_2 branch for input $\hat{\mathbf{x}}_i$, respectively.

And the gradients of the Eq. (9) is computed as follow.

$$\begin{aligned} \nabla L_{aug-supervised} = & \sum_{i=1, j=r(i)}^{B^l} \{[\lambda(\rho(\hat{\mathbf{x}}_i^l, \theta) - \mathbf{y}_i^*) \\ & + (1 - \lambda)(\rho(\hat{\mathbf{x}}_i^l, \theta) - \mathbf{y}_j^*)] \cdot \nabla \rho(\hat{\mathbf{x}}_i^l, \theta)\} \end{aligned} \quad (13)$$

where $\nabla \rho(\hat{\mathbf{x}}_i^l, \theta)$ is the gradient of softmax probability $\rho(\hat{\mathbf{x}}_i^l, \theta)$.

Finally, we minimize Eq. (1) and update the network parameters with the gradients. Through Algorithm 1, the proposed network of SDB-Net can be learned efficiently. The SDB-Net addresses the learned feature distribution mismatch problem by separating the effects caused by the ground truth of labeled data and the inferred labels of unlabeled data through the dual branch architecture. The co-consistency loss encourages the C_2 branch to learn the matching distribution for labeled and unlabeled data by using predictions of unlabeled data only from the C_1 branch. Then the shared module M makes the feedback to the C_1 branch to complete the learning cycle.

4. Experiments

In this section, we first introduce the benchmark datasets used in our evaluation experiments, followed by the experimental setup. Then, we conduct a feature visualization experiment to show how the SDB-Net works and compare our method with the related works, including the state-of-the-art semi-supervised methods. Finally, we perform an ablation study to show the impacts of different loss terms involved in the proposed method.

4.1. Experimental setup

Three popular image classification datasets, SVHN [45], CIFAR-10 and CIFAR-100 [46], are used to evaluate the performances of our method. Following the standard practice in the semi-supervised literatures [31,30,32,47], we apply the standard data-augmentation and pre-processing schemes to the benchmarks. Mean error rate and standard deviation are reported in our experiments by averaging over 5 runs with different seeds for data splits.

Algorithm 1 Algorithm of optimizing SDB-Net.

Require: A mini-batch of labeled samples $\mathbf{X}^l$ and their ground truths $\mathbf{Y}^l$, a mini-batch of unlabeled samples $\mathbf{X}^u$.

1. Initialize the network parameters θ , w_1 , w_2 , α .

repeat

2. Perform a forward pass on $\mathbf{X}^u$ to compute the co-consistency loss via Eq. (2).

3. Sample λ from $\beta(\alpha, \alpha)$ to generate mixup labeled data $\hat{\mathbf{X}}^l = \text{Mixup}(\mathbf{X}^l)$ via Eq. (6).

4. Perform a forward pass in C_1 branch on $\hat{\mathbf{X}}^l$ to compute the augmentation supervised loss via Eq. (9).

5. Let $\mathbf{X} = \text{Concat}(\mathbf{X}^l, \mathbf{X}^u)$ and apply Mixup on $\mathbf{X}$ to generate mixup data $\{(\hat{\mathbf{x}}, \hat{\mathbf{y}}^1, \hat{\mathbf{y}}^2)\}$ via Eq. (3).

6. Perform a forward pass on $\{(\hat{\mathbf{x}}, \hat{\mathbf{y}}^1, \hat{\mathbf{y}}^2)\}$ to compute the mix-consistency regularizations via Eqs. (4) and (5).

7. Update the network via mini-batch SGD by minimizing Eq. (1) based on the gradients of Eqs. (10), (11), (12), and (13).

until the maximum number of iterations is reached.

return The network parameters θ .

SVHN: The Street View House Numbers (SVHN) dataset contains 73,257 training samples and 26,032 testing samples belonging to 10 different classes. Each sample is a RGB image of a real-world house number. For SVHN, we use the zero-padding in each image with 2 pixels and randomly cropping to generate a new image and then normalize images to the ones with zero mean and unit variance. The image size for all images is 32×32 .

CIFAR-10: CIFAR-10 dataset consists of RGB images with the size of 32×32 and contains 50,000 training samples and 10,000 testing samples. CIFAR-10 has 10 different classes, which include photos of natural objects such as cars, horses, airplanes and deer. For CIFAR-10, we first zero-pad each sample with 2 pixels, randomly crop the image with the resolution of 32×32 , and then horizontally flip the resulting images. Next, the zero-mean and ZCA whitening [46] are applied to these images.

CIFAR-100: Similarly to CIFAR-10, CIFAR-100 has 50,000 training samples and 10,000 testing samples with resolution of 32×32 , coming from 100 different classes. We follow a standardized procedure equivalent to the one of CIFAR-10 except ZCA whitening.

4.2. Parameter discussion

In the semi-supervised experiments, we use a momentum optimizer to train the proposed SDB-Net. We set the momentum as 0.9. Then we run the experiments for 400 epochs in total and the hyperparameters w_1 and w_2 ramp up their weights from 0 to the final values during the first 80 epochs. The initial learning rate is set as 0.1 and then annealed by a cosine annealing schedule proposed in [48]. The weight decay coefficient is set as 0.0001 and the batch-size is 100 in our experiments. The network backbone is a 13-layer convolution network (i.e., CNN-13 architecture shown in Table 1) following the previous works [31,32,30,47]. We remove the additive Gaussian noise in the input layer and the Dropout noise for focusing on the advantages of our method.

We conduct a grid search to select the best hyperparameters by using a validation set of {5000, 10000, 1000} labeled samples for CIFAR-10, CIFAR-100 and SVHN, respectively. In inference, given a test sample, we use the average of the posterior probability of the two branch modules as the final prediction output of our model.

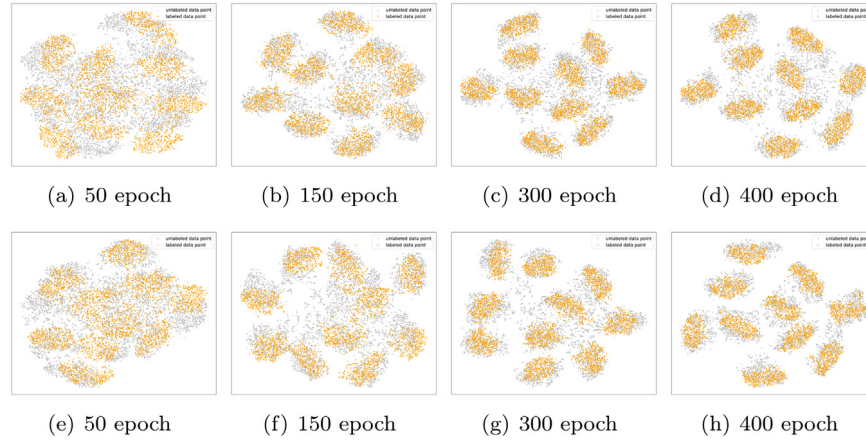


Fig. 4. The evolution of the feature representations of SDB-Net. The top(a-d) are the features from C_1 branch and the bottom(e-h) are from C_2 branch. The orange points denote labeled data points and the silver points denote unlabeled data. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4.3. Feature visualization

In this subsection, we compare the final feature representation distributions between our method and the semi-supervised baseline, Pseudo-Label [28], with the base CNN-13 architecture as a feature extractor. Then we further visualize the evolution of the feature representations of our SDB-Net under the same setting. The experiment is conducted on CIFAR-10 dataset. We use 4000 randomly selected training samples as labeled data, with the remaining 46,000 training samples as unlabeled data. We train the Pseudo-Label and SDB-Net in the same experimental setup mentioned in Section 4.2. For better visualization, the resulting 2-D deep features are generated by t-SNE [34] approach for 4000 labeled training samples and 4000 randomly selected unlabeled training samples.

The labeled features and unlabeled features of Pseudo-Label are shown in (a), (b), and (c) of Fig. 1. It is observed that the learned feature distribution of labeled data mismatches that of unlabeled data. On one hand, the labeled features in Fig. 1(a) clearly form several clusters and there is not obvious separation in the unlabeled features as shown in Fig. 1(b). On the other hand, from Fig. 1(c), though unlabeled features have some topological structures, the separation of labeled features does not match with that of the unlabeled features. Although the Pseudo-Label can employ the unlabeled data by regarding the predicted classes of unlabeled data as their labels, the feature distribution of unlabeled data is tangled and inconsistent with that of labeled data. Moreover, the concentrated labeled data clusters show that the Pseudo-Label suffers from the overfitting due to the limited labeled data.

By contrast, the final feature distributions of labeled and unlabeled data from SDB-Net shown in (d) and (h) of Fig. 4 are matching in both two branches and SDB-Net less suffers from the overfitting. As shown in Fig. 4, with our SDB-Net, the learned feature distributions of labeled and unlabeled data gradually form 10 separable clusters in the training procedure and eventually the clusters of labeled feature and clusters of unlabeled feature match each other.

Besides, we design a center distance to measure the distance between the learned labeled features and unlabeled features. Let $(\mathbf{z}_i^l, \mathbf{y}_i^l) \in \{\mathbf{Z}^l, \mathbf{Y}^l\}$ be the labeled feature representations, $\mathbf{z}_j^u \in \mathbf{Z}^u$ be the unlabeled feature representations. Denote A the number of classes, then the dataset of labeled and unlabeled data can be split into A subsets $\{S_1, \dots, S_k, \dots, S_A\}$ based on the ground truth.

The center distance can be defined as follow.

$$d_{center} = \frac{1}{A} \sum_{k=1}^A \|\bar{\mathbf{c}}_k^l - \bar{\mathbf{c}}_k^u\|^2 \quad (14)$$

where $\bar{\mathbf{c}}_k^l = \frac{1}{N_k^l} \sum_{\mathbf{z}_i^l \in S_k} \mathbf{z}_i^l$ and N_k^l is the number of labeled samples in S_k . For unlabeled data, $\bar{\mathbf{c}}_k^u = \frac{1}{N_k^u} \sum_{\mathbf{z}_j^u \in S_k} \mathbf{z}_j^u$ and N_k^u is the number of unlabeled samples in S_k . The results are shown in Fig. 5(a).

From Fig. 5(a), we can observe that in the beginning of Pseudo-Label's training, the distance between labeled features and unlabeled features gradually decreases. However, due to the learned feature distribution mismatch problem and overfitting, the distance between labeled features and unlabeled features increases in the later training processing. And the center distance of our SDB-Net also decreases in the beginning of training in general. With the dual-branch architecture and co-consistency regularization, our SDB-Net keeps the learned feature representations matching in the later training.

We show the train losses and test classification accuracies of our SDB-Net and Pseudo-Label during training in Fig. 5. From Figs. 5(b) and 5(c), we can observe that the train loss of Pseudo-Label decreases quickly in the beginning of training and later stays in a low value which makes slow progress of Pseudo-Label. Due to the strong stochastic perturbations injected by MixUp, train loss of SDB-Net seems unstable while constantly decreases. With the mix-consistency regularization and co-consistency regularization, the performance of our SDB-Net improves continuously as the training goes on. In Fig. 6, we also display some test samples which have been wrongly recognized by Pseudo-Label but correctly classified by SDB-Net. While the samples in automobile class are often incorrectly classified to truck class and samples in deer class are often incorrectly classified as horse or dog by Pseudo-Label, they can be correctly recognized by SDB-Net.

4.4. Comparison with the state-of-the-arts

In this experiment, we compare our model with recently proposed state-of-the-art semi-supervised deep learning approaches, including consistency-based methods (e.g., MeanTeacher [30], Π model and Tempens [31]), graph-based methods (e.g., Π + SNTG [38], LP-SSDL [39] and its variant, LP-SSDL-MT), meanteacher-based variants (e.g., Filtering CCL and Temperature CCL [36],

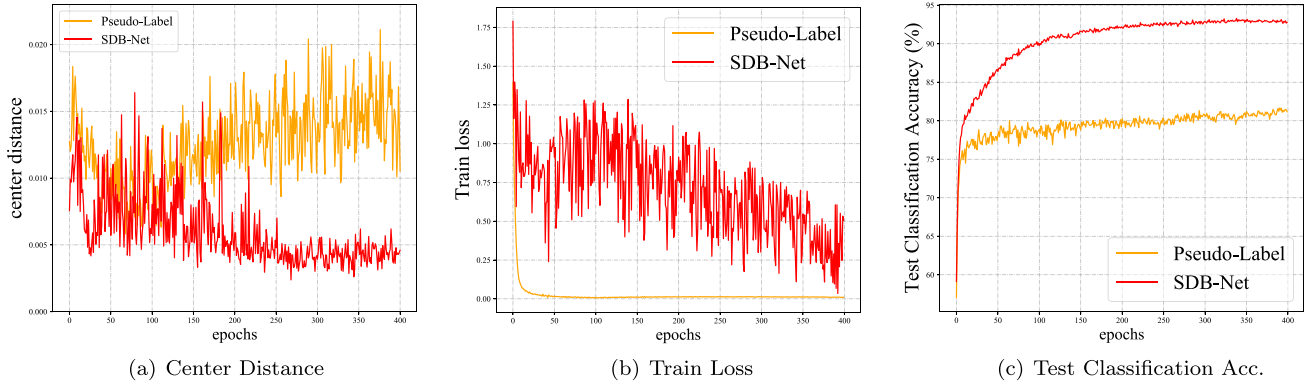


Fig. 5. The comparative analysis images of our SDB-Net and a baseline method, Pseudo-Label.

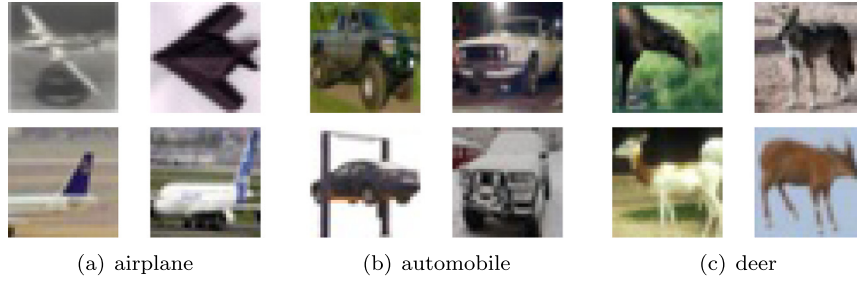


Fig. 6. Some test samples from airplane, automobile and deer classes in CIFAR-10 dataset. The test samples have been wrongly recognized by Pseudo-Label but correctly classified by SDB-Net.

TSSDL [40] and its variant, TSSDL-MT, and ICT [37]), TNAR-VAE [41] and MA-DNN [29]. We also compare against the supervised model with labeled data only to justify the improvement of our method.

For CIFAR-10, we evaluate the error rate of each method by randomly selecting {100, 200, 400} labeled samples from each class in the train set as the labeled data, with the remaining of train set as unlabeled data. For CIFAR-100 and SVHN, we randomly select 100 images from each class as labeled data.

The experiment is conducted on all three datasets and the results are shown in Tables 2 and 3. The comparison includes performances reported in prior literatures, our reproduced results of Π model [31] and Tempens [31] in 100-labels and 200-labels CIFAR-10 experiments, and ICT [37] in CIFAR-100 experiments are marked by “*”.

As shown in Tables 2 and 3, the proposed SDB-Net substantially improves the supervised counterpart and achieves the best performance in SVHN and CIFAR-100 experiments and beats most start-of-the-art methods on CIFAR-10. In the CIFAR-10 100-label experiment, our SDB-Net achieves an error rate of 11.91%, which significantly outperforms the error rate of 15.48% obtained by the next-best-performing method (ICT). Most other semi-supervised methods have used the Dropout noise, which gives them more regularizing capabilities. Despite this, our method clearly outperforms other methods except the experimental result of ICT in the CIFAR-10 400-label case. The best performances achieved by the proposed SDB-Net in the experiments with very limited labeled data (e.g., 100-labels and 200-labels CIFAR-10 experiments, 100-labels CIFAR-100 experiment and 100-labels SVHN experiment), show that our method handles well the overfitting due to labeled data and the learned feature distribution mismatch problem.

4.5. Comparison with the related works

We further present a comparison with recent related works, including split-BN [43], LGA [44] and ADA-Net [42], which implicitly alleviate the learned feature distribution mismatch issue of

Table 2

Error rate (%) of comparison of the start-of-the-arts on CIFAR-10. The results marked ‘-’ have not been reported in literatures.

Method	100	200	400
Π model [31]	31.65 $\pm$ 1.20*	17.57 $\pm$ 0.44*	12.36 $\pm$ 0.31
Tempens [31]	23.31 $\pm$ 1.01*	15.64 $\pm$ 0.39*	12.16 $\pm$ 0.24
MeanTeacher [30]	21.55 $\pm$ 1.48	15.73 $\pm$ 0.31	12.31 $\pm$ 0.28
TSSDL [40]	21.13 $\pm$ 1.17	14.65 $\pm$ 0.33	10.90 $\pm$ 0.23
TSSDL-MT [40]	18.41 $\pm$ 0.92	13.54 $\pm$ 0.32	9.30 $\pm$ 0.55
Π + SNTG [38]	21.23 $\pm$ 1.27	14.65 $\pm$ 0.31	11.36 $\pm$ 0.31
Filtering CCL [36]	16.99 $\pm$ 0.71	12.57 $\pm$ 0.47	10.63 $\pm$ 0.22
Temperature CCL [36]	17.26 $\pm$ 0.69	12.45 $\pm$ 0.33	10.73 $\pm$ 0.26
LP-SSDL [39]	22.02 $\pm$ 0.88	15.66 $\pm$ 0.35	12.69 $\pm$ 0.29
LP-SSDL-MT [39]	16.93 $\pm$ 0.70	13.22 $\pm$ 0.29	10.61 $\pm$ 0.28
ICT [37]	15.48 $\pm$ 0.78	9.26 $\pm$ 0.09	7.29 $\pm$ 0.02
TNAR-VAE [41]	-	-	8.85 $\pm$ 0.03
Supervised only	37.61 $\pm$ 0.92	28.92 $\pm$ 0.47	23.70 $\pm$ 0.88
SDB-Net	11.91 $\pm$ 0.27	9.03 $\pm$ 0.22	7.60 $\pm$ 0.13

Table 3

Error rate (%) of comparison of the start-of-the-arts on CIFAR-100 and SVHN. The results marked ‘-’ have not been reported in literatures.

Method	CIFAR-100	SVHN
Π model [31]	39.19 $\pm$ 0.36	4.82 $\pm$ 0.17
Tempens [31]	38.65 $\pm$ 0.51	4.42 $\pm$ 0.16
MeanTeacher [30]	36.08 $\pm$ 0.53	3.95 $\pm$ 0.19
MA-DNN [29]	36.08 $\pm$ 0.53	3.95 $\pm$ 0.19
Π + SNTG [38]	37.97 $\pm$ 0.29	3.82 $\pm$ 0.25
Filtering CCL [36]	34.81 $\pm$ 0.52	3.86 $\pm$ 0.19
Temperature CCL [36]	35.15 $\pm$ 0.62	3.93 $\pm$ 0.21
ICT [37]	33.46 $\pm$ 0.32*	3.93 $\pm$ 0.21
LP-SSDL [39]	38.43 $\pm$ 1.88	-
LP-SSDL-MT [39]	35.92 $\pm$ 0.47	-
TNAR-VAE [41]	-	3.74 $\pm$ 0.04
Supervised only	42.68 $\pm$ 0.51	13.12 $\pm$ 0.95
SDB-Net	31.32 $\pm$ 0.67	3.72 $\pm$ 0.06

Table 4

Error rate (%) of comparison with related works on CIFAR-10 and SVHN. The results marked ‘-’ have not been reported in literatures.

Method	CIFAR-10	SVHN
Split-BN [43]	14.8 ± 0.9	-
LGA [44]	12.91 ± 0.15	7.56 ± 0.19
LGA + VAT [44]	12.06 ± 0.19	6.58 ± 0.36
ADA-Net [42]	10.3	4.62
Supervised only	23.70 ± 0.88	13.12 ± 0.95
SDB-Net	7.79±0.13	3.72±0.06

labeled and unlabeled data. We report the comparison results of different methods on the CIFAR-10 with 400 labeled samples per class and SVHN with 100 labeled samples per class in Table 4. And for a fair comparison with ADA-Net, we remove the operation of data augmentation except zero-mean in CIFAR-10 experiment of our method.

From Table 4, we can see that the proposed SDB-Net beats all related methods and achieves the best performances in both CIFAR-10 and SVHN experiments. Especially in complex natural image dataset CIFAR-10, our SDB-Net can exploit more structure information from unlabeled data and significantly outperform all other methods in Table 4. Comparing with the related methods that use sophisticated architectures (e.g., ADA-Net [42]) and complex computing metrics (e.g., LGA and LGA + VAT [44]), the proposed SDB-Net is more effective to solve the learned feature distribution mismatch problem. With encouraging the learned feature distributions of labeled and unlabeled data to match, our dual branch architecture can efficiently boost the model performance.

4.6. Varying number of labeled samples

In this section, we conduct the experiments to validate the effectiveness of our SDB-Net by varying the number of labeled samples on CIFAR-10, CIFAR-100, and SVHN datasets. $\mathcal{I}$ model [31] and MeanTeacher [30] are used as baselines. Specifically, we randomly select {50, 100, 200, 400}, {20, 40, 100} and {25, 50, 100} from each class on CIFAR-10, CIFAR-100 and SVHN, respectively. We set the hyperparameter w_2 to 1.0 since it works well in all experiments in the validation sets. For other hyperparameters w_1 and α , we search them by starting from similar experimental settings in previous works [30,37]. We set $w_1 = 100$ that works well in all experiments. The hyperparameter α in $\beta(\alpha, \alpha)$ is set to {0.1, 0.2, 0.9, 0.9} in the experiments in CIFAR-10 with {50, 100, 200, 400} labeled samples. For the experiments on CIFAR-100 and SVHN, we set $\alpha = 0.1$. In brief, for less number of labeled data, lower value of α works well.

The error rates of our SDB-Net and the baselines are plotted in Fig. 7. We can observe that SDB-Net consistently outperforms the competitive methods and improves the classification performances in all experiments with different number of labeled samples. Especially in 50-labels and 100-labels CIFAR-10 experiments and 20-labels CIFAR-100 experiment, the proposed SDB-Net shows remarkable improvements and significantly surpasses other competitive methods. The results of experiments with very limited labeled data indicate that the dual branch architecture and augmentation supervised loss enable our method to effectively alleviate the overfitting and significantly boost the performance.

4.7. Ablation study

Our SDB-Net consists of three loss terms, i.e., co-consistency loss, mix-consistency regularization, and augmentation supervised loss. In this experiment, we investigate the effects of removing the loss terms to obtain the additional insight into how the SDB-Net works. Specifically, we study the impact of

Table 5

Error rate (%) on ablation study.

	-o co	-r mix	-r aug	100	400
Supervised only				37.61	23.7
SDB-Net				11.91	7.60
✓				15.02	11.17
		✓		13.36	10.54
			✓	12.58	7.99
✓	✓	✓		19.98	12.19
✓	✓		✓	16.69	11.35
		✓	✓	15.17	10.79

Table 6

The execution time (seconds) of our SDB-Net and competitive methods.

	$\mathcal{I}$ model	MeanTeacher	ICT	SDB-Net
Train iteration time	0.127	0.218	0.444	0.497
Inference iteration time	0.092	0.097	0.095	0.109

- -o co: remove the dual branch architecture and co-consistency loss;
- -r mix: replace the mix-consistency regularization with the consistency regularization from [30];
- -r aug: replace the augmentation supervised loss with a plain supervised loss using cross-entropy cost function.

We conduct the ablation study on the CIFAR-10 dataset with {100, 400} labeled samples per class. The results are shown in Table 5, from which we observe that each term contributes to boosting the performance. In the “-o co” and “-r mix” 100-label CIFAR-10 experiments, the proposed augmentation supervised loss reduces the classification error rate to a nice performance 19.98% and shows the effectiveness of alleviating overfitting due to the limited labeled data. In the “-o co” and “-r aug” 100-labels CIFAR-10 experiments, the SDB-Net with only mix-consistency regularization achieves the error rate of 16.69% and outperforms most semi-supervised methods in Table 2. In the “-r mix” and “-r aug” experiments, the good performances of 15.17% and 10.79% demonstrate the effectiveness of our dual-branch architecture and the proposed co-consistency loss.

4.8. Execution time

In this section, we conduct the experiment to investigate the execution time. We report the train iteration time (i.e., the time cost of running a mini-batch of data) and inference iteration time, by averaging the iteration time in first epoch. We evaluate the execution time in CIFAR-10 dataset with 1000 randomly selected training samples as labeled data. The batch size is set to 100. And we run the public codes¹²³ to obtain the iteration times of $\mathcal{I}$ model [31], MeanTeacher [30] and ICT [37] in the same setting for comparison. The experiment is performed on a GTX 1080Ti GPU with Pytorch-0.4.1 software toolkit. The system memory is 251 GB and CPU is Intel Xeon E5-2650 v4. The experimental results are shown in Table 6 and Fig. 8.

From Table 6 and Fig. 8, $\mathcal{I}$ model has lowest train iteration time and lowest test classification accuracy. Due to weight averaging and MixUp, ICT costs more training time than $\mathcal{I}$ model and MeanTeacher, but has higher classification accuracy. With efficient implementing of dual-branch network and co-consistency regularization, the proposed SDB-Net achieves higher classification accuracy than ICT and costs a little extra computational

¹ <https://github.com/smlaine2/tempens>

² <https://github.com/CuriousAI/mean-teacher>

³ <https://github.com/vikasverma1077/ICT>

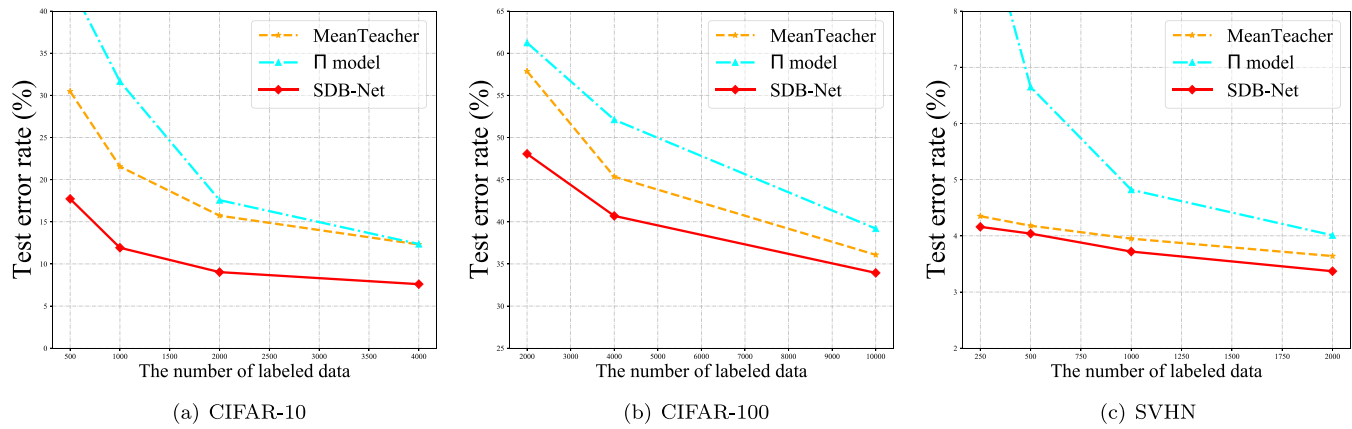


Fig. 7. Test error rate of our SDB-Net and baselines methods, Π model and MeanTeacher, as the amount of labeled data varies.

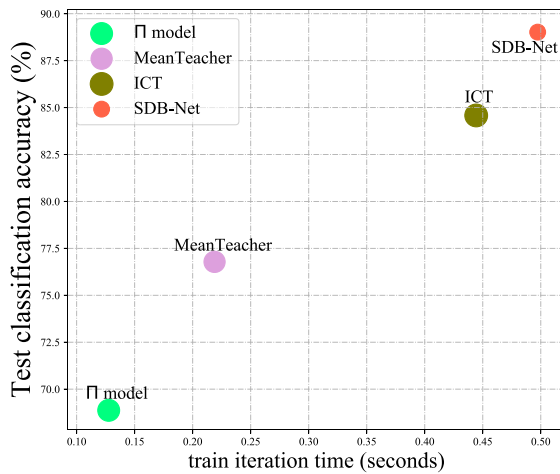


Fig. 8. The train iteration time (seconds) and test classification accuracies of our SDB-Net and competitive methods, Π model, MeanTeacher and ICT. The different sizes of color points show the standard deviations. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

burden. In inference iteration time in Table 6, due to the same network architecture backbone, the time costs of all methods are similar. Although our SDB-Net has two branches, the time cost of branch module is negligible compared to the whole deep network.

5. Conclusion

In this paper, we first revealed a rarely discussed problem about learned feature distribution mismatch between labeled data and unlabeled data, which hurts the semi-supervised smoothness assumption. To tackle the mismatch problem, we proposed a new semi-supervised learning method, called semi-supervised Dual-Branch Network (SDB-Net). In particular, we designed a novel dual branch architecture and a simple yet effective co-consistency loss, which have well alleviated the mismatch issue by using inferred labels of unlabeled data only. We have also designed a consistency regularization variant, mix-consistency regularization, and an augmentation supervised loss, which have performed well in preventing from overfitting and further relieving the mismatch problem. The effectiveness of the proposed SDB-Net has been successfully demonstrated on three benchmarks, CIFAR-10, CIFAR-100 and SVHN, on which our method outperforms the state-of-the-art methods.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Jiaming Chen: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing - original draft, Visualization. **Meng Yang:** Conceptualization, Methodology, Validation, Formal analysis, Investigation, Resources, Data curation, Writing - original draft, Writing - review & editing, Supervision, Project administration, Funding acquisition. **Guangwei Gao:** Writing - review & editing, Project administration, Funding acquisition.

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References

- [1] O. Russakovsky, J. Deng, H. Su, J. Krause, S. Satheesh, S. Ma, Z. Huang, A. Karpathy, A. Khosla, M. Bernstein, A.C. Berg, L. Fei-Fei, Imagenet large scale visual recognition challenge, *Int. J. Comput. Vis.* 115 (3) (2015) 211–252, <http://dx.doi.org/10.1007/s11263-015-0816-y>.
- [2] T.-Y. Lin, P. Goyal, R. Girshick, K. He, P. Dollar, Focal loss for dense object detection, in: *The IEEE International Conference on Computer Vision, ICCV*, 2017, pp. 2999–3007.
- [3] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: *2016 IEEE Conference on Computer Vision and Pattern Recognition, CVPR*, 2016, pp. 770–778, <http://dx.doi.org/10.1109/CVPR.2016.90>.
- [4] K. He, X. Zhang, S. Ren, J. Sun, Identity mappings in deep residual networks, in: *The European Conference on Computer Vision, ECCV*, Springer International Publishing, 2016, pp. 630–645.
- [5] G. Huang, Z. Liu, L. v.d. Maaten, K.Q. Weinberger, Densely connected convolutional networks, in: *2017 IEEE Conference on Computer Vision and Pattern Recognition, CVPR*, 2017, pp. 2261–2269, <http://dx.doi.org/10.1109/CVPR.2017.243>.
- [6] M. Ren, S. Ravi, E. Triantafyllou, J. Snell, K. Swersky, J.B. Tenenbaum, H. Larochelle, R.S. Zemel, Meta-learning for semi-supervised few-shot classification, in: *International Conference on Learning Representations*, 2018.

- [7] V.G. Satorras, J.B. Estrach, Few-shot learning with graph neural networks, in: International Conference on Learning Representations, 2018.
- [8] G. Balakrishnan, A. Zhao, M.R. Sabuncu, A.V. Dalca, J. Guttag, An unsupervised learning model for deformable medical image registration, in: 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2018, pp. 9252–9260, <http://dx.doi.org/10.1109/CVPR.2018.00964>.
- [9] A.V. Dalca, J. Guttag, M.R. Sabuncu, Anatomical priors in convolutional networks for unsupervised biomedical segmentation, in: 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2018, pp. 9290–9299, <http://dx.doi.org/10.1109/CVPR.2018.00968>.
- [10] Z. Zhang, Y. Xie, F. Xing, M. McGough, L. Yang, Mdnets: A semantically and visually interpretable medical image diagnosis network, in: 2017 IEEE Conference on Computer Vision and Pattern Recognition, CVPR, 2017, pp. 3549–3557, <http://dx.doi.org/10.1109/CVPR.2017.378>.
- [11] O. Chapelle, B. Schölkopf, A. Zien, *Semi-Supervised Learning*, MIT Press, 2006.
- [12] K. Liu, X. Yang, H. Yu, J. Mi, P. Wang, X. Chen, Rough set based semi-supervised feature selection via ensemble selector, *Knowl.-Based Syst.* 165 (2019) 282–296.
- [13] T. Joachims, Transductive inference for text classification using support vector machines, in: Proceedings of the Sixteenth International Conference on Machine Learning, ICML '99, Morgan Kaufmann Publishers Inc., San Francisco, CA, USA, 1999, pp. 200–209.
- [14] K.P. Bennett, A. Demiriz, Semi-supervised support vector machines, in: Proceedings of the 1998 Conference on Advances in Neural Information Processing Systems II, MIT Press, Cambridge, MA, USA, 1999, pp. 368–374.
- [15] R.R. Khemchandani, A. Pal, Fuzzy semi-supervised weighted linear loss twin support vector clustering, *Knowl.-Based Syst.* 165 (2019) 132–148.
- [16] A. Blum, T. Mitchell, Combining labeled and unlabeled data with co-training, in: Proceedings of the Eleventh Annual Conference on Computational Learning Theory, 1998, pp. 92–100, <http://dx.doi.org/10.1145/279943.279962>.
- [17] A. Blum, S. Chawla, Learning from labeled and unlabeled data using graph mincuts, in: Proceedings of the Eighteenth International Conference on Machine Learning, ICML '01, Morgan Kaufmann Publishers Inc., San Francisco, CA, USA, 2001, pp. 19–26.
- [18] B. Schölkopf, J. Platt, T. Hofmann, *Learning on Graph with Laplacian Regularization*, MIT Press, 2007.
- [19] O. Chapelle, J. Weston, B. Schölkopf, Cluster kernels for semi-supervised learning, in: Proceedings of the 15th International Conference on Neural Information Processing Systems, NIPS'02, MIT Press, Cambridge, MA, USA, 2002, pp. 601–608.
- [20] J. Steffen, S. Klanke, S. Vijayakumar, H. Ritter, Towards semi-supervised manifold learning: Ukr with structural hints, in: J.C. Príncipe, R. Miikkilainen (Eds.), *Advances in Self-Organizing Maps*, Springer Berlin Heidelberg, Berlin, Heidelberg, 2009, pp. 298–306.
- [21] R.K. Sanodiya, J. Mathew, A framework for semi-supervised metric transfer learning on manifolds, *Knowl.-Based Syst.* 176 (2019) 1–14.
- [22] H. Wu, S. Prasad, Semi-supervised deep learning using pseudo labels for hyperspectral image classification, *IEEE Trans. Image Process.* 27 (3) (2018) 1259–1270, <http://dx.doi.org/10.1109/TIP.2017.2772836>.
- [23] Y. Kuznetsov, J. Stückler, B. Leibe, Semi-supervised deep learning for monocular depth map prediction, in: 2017 IEEE Conference on Computer Vision and Pattern Recognition, CVPR, 2017, pp. 2215–2223, <http://dx.doi.org/10.1109/CVPR.2017.238>.
- [24] A. Rasmus, M. Berglund, M. Honkala, H. Valpola, T. Raiko, Semi-supervised learning with ladder networks, in: C. Cortes, N.D. Lawrence, D.D. Lee, M. Sugiyama, R. Garnett (Eds.), *Advances in Neural Information Processing Systems*, vol. 28, Curran Associates, Inc., 2015, pp. 3546–3554.
- [25] S. Rifai, Y.N. Dauphin, P. Vincent, Y. Bengio, X. Muller, The manifold tangent classifier, in: International Conference on Neural Information Processing Systems, 2011, pp. 2294–2302.
- [26] Y. Cheng, X. Zhao, R. Cai, Z. Li, K. Huang, Y. Rui, Semi-supervised multimodal deep learning for rgb-d object recognition, in: Proceedings of the Twenty-Fifth International Joint Conference on Artificial Intelligence, IJCAI'16, AAAI Press, 2016, pp. 3345–3351.
- [27] H. Wu, S. Prasad, Semi-supervised deep learning using pseudo labels for hyperspectral image classification, *IEEE Trans. Image Process.* 27 (3) (2018) 1259–1270, <http://dx.doi.org/10.1109/TIP.2017.2772836>.
- [28] D.-H. Lee, Pseudo-label: The simple and efficient semi-supervised learning method for deep neural networks, in: ICML 2013 Workshop : Challenges in Representation Learning, WREPL, 2013.
- [29] Y. Chen, X. Zhu, S. Gong, Semi-supervised deep learning with memory, in: The European Conference on Computer Vision, ECCV, 2018.
- [30] A. Tarvainen, H. Valpola, Mean teachers are better role models: Weight-averaged consistency targets improve semisupervised deep learning results, in: *Advances in Neural Information Processing Systems*, 2017, pp. 1195–1204.
- [31] S. Laine, T. Aila, Temporal ensembling for semisupervised learning, in: Proceedings of International Conference on Learning Representations, ICLR, 2017.
- [32] T. Miyato, S. Maeda, S. Ishii, M. Koyama, Virtual adversarial training: A regularization method for supervised and semi-supervised learning, *IEEE Trans. Pattern Anal. Mach. Intell.* 41 (8) (2018) 1979–1993, <http://dx.doi.org/10.1109/TPAMI.2018.2858821>.
- [33] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, R. Salakhutdinov, Dropout: a simple way to prevent neural networks from overfitting, *J. Mach. Learn. Res.* 15 (1) (2014) 1929–1958.
- [34] L. van der Maaten, G. Hinton, Visualizing data using t-SNE, *J. Mach. Learn. Res.* 9 (2008) 2579–2605.
- [35] H. Zhang, M. Cisse, Y.N. Dauphin, D. Lopez-Paz, mixup: Beyond empirical risk minimization, in: International Conference on Learning Representations, 2018.
- [36] Y. Li, L. Liu, R.T. Tan, Certainty-driven consistency loss for semi-supervised learning, *CoRR abs/1901.05657*, [arXiv:1901.05657](https://arxiv.org/abs/1901.05657).
- [37] V. Verma, A. Lamb, J. Kannala, Y. Bengio, D. Lopez-Paz, Interpolation consistency training for semi-supervised learning, in: Proceedings of International Joint Conference on Artificial Intelligence, IJCAI, 2019.
- [38] Y. Luo, J. Zhu, M. Li, Y. Ren, B. Zhang, Smooth neighbors on teacher graphs for semi-supervised learning, in: 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2018, pp. 8896–8905, <http://dx.doi.org/10.1109/CVPR.2018.00927>.
- [39] A. Iscen, G. Tolias, Y. Avrithis, O. Chum, Label propagation for deep semi-supervised learning, in: The IEEE Conference on Computer Vision and Pattern Recognition, CVPR, 2019.
- [40] W. Shi, Y. Gong, C. Ding, Z. MaXiaoyu Tao, N. Zheng, Transductive semi-supervised deep learning using min-max features, in: The European Conference on Computer Vision, ECCV, 2018.
- [41] B. Yu, J. Wu, J. Ma, Z. Zhu, Tangent-normal adversarial regularization for semi-supervised learning, in: The IEEE Conference on Computer Vision and Pattern Recognition, CVPR, 2019.
- [42] Q. Wang, W. Li, L.V. Gool, Semi-supervised learning by augmented distribution alignment, in: The IEEE International Conference on Computer Vision, ICCV, 2019.
- [43] M. Zajac, K. Żołna, S. Jastrzębski, Split batch normalization: a simple trick for semi-supervised learning under domain shift, in: ICLR Workshop Learning from Limited Labeled Data, LLD, 2019.
- [44] J. Jackson, J. Schulman, Semi-supervised learning by label gradient alignment, *CoRR abs/1902.02336*, [arXiv:1902.02336](https://arxiv.org/abs/1902.02336).
- [45] Y. Netzer, T. Wang, A. Coates, A. Bissacco, B. Wu, A.Y. Ng, Reading digits in natural images with unsupervised feature learning, *NIPS Workshop on Deep Learning and Unsupervised Feature Learning*.
- [46] A. Krizhevsky, G. Hinton, Learning Multiple Layers of Features from Tiny Images, Tech. Rep 1, Computer Science Department, University of Toronto.
- [47] A. Oliver, A. Odena, C.A. Raffel, E.D. Cubuk, I. Goodfellow, Realistic Evaluation of Deep Semi-Supervised Learning Algorithms, in: *Advances in Neural Information Processing Systems*, vol. 31, Curran Associates, Inc., 2018, pp. 3235–3246.
- [48] I. Loshchilov, F. Hutter, SGDR: Stochastic gradient descent with restarts, in: Proceedings of International Conference on Learning Representations, ICLR, 2017.